A FOUR-VECTOR CONTROL ARCHITECTURE FOR VIABILITY IN BIOLOGICAL AND SYNTHETIC AGENTS

UTSM: A persistent motivational ordering of Agency, Attunement, Inference, and Stability

Agency • Attunement • Inference • Stability

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Abstract

This manuscript proposes the Unified Theory of Subconscious Motivation (UTSM): a viability-first control architecture for time-persistent, model-updating, goal-selecting agents. UTSM models subconscious motivation as four continuously active viability vectors — Agency, Attunement, Inference, and Stability — organised as a persistent motivational ordering (a 'motivational stack' in this paper's shorthand). Claims are made at three levels: terms and scope conditions are stipulative; architectural claims about ordering, gain, and constraint-binding are theoretical proposals; and empirical claims are expressed as falsifiable predictions rather than proof. Vectors are defined by the classes of viability error they regulate — efficacy/controllability, coupling/relevance, model/coherence, and integrity/recovery — rather than by surface emotions, traits, or domain-specific objects of desire. Object-level biological drives (e.g., hunger or mating) are treated as specialised error signals and goal contents that recruit, and are constrained by, this four-vector arbitration rather than as additional fundamental controllers. The framework separates trait-like vector rank (persistent ordering) from state-like vector gain (context-sensitive excitability driven by discrepancy between perceived and desired situation/self context) and predicts ordered co-activation: in dyads, the dominant vector specifies the end-state objective while the secondary vector specifies method; remaining vectors act as constraints and veto logic. UTSM is framed as a conditional Minimum Viable Agent hypothesis: within scope, long-run coherent thought and goal pursuit require closure of all four control loops. The hypothesis is falsifiable via selective ablation in synthetic agents (or approximate suppression in humans): sustained removal of any one vector should yield a characteristic failure phenotype. Meaning is operationalised geometrically: motivational state modulates similarity for attention and retrieval, so that proximity becomes relevance; drift corresponds to measurable deviations in motivational geometry. The paper derives feasibility constraints for persistent looped agents, proposes drive-aware diagnostics, and outlines an empirical programme spanning behavioural tasks, physiology, neuroimaging, and synthetic ablations, with applications to personality modelling, clinical hypothesis generation (non-diagnostic), organisational design, and AI alignment.

Keywords: subconscious motivation; motivational ordering; viability vectors; Agency; Attunement; Inference; Stability; cognitive architecture; control theory; predictive processing; synthetic agents; AI alignment; active inference.

1. Introduction

Motivation is often treated as a derivative phenomenon: a downstream expression of affect, values, reinforcement history, or deliberate reasoning. These framings can describe behaviour, but they struggle to explain why behaviour remains goal-directed, stable across time, and self-consistent across contexts — especially in agents that must persist, learn, and act under uncertainty. Across psychology, neuroscience, and artificial intelligence, we can often model what an agent does; we less often model the compact control architecture that makes an agent keep doing it coherently. This manuscript proposes the Unified Theory of Subconscious Motivation (UTSM): a minimal, cross-domain architecture in which subconscious motivation is modelled as four continuous viability vectors — Agency, Attunement, Inference, and Stability — organised as a persistent motivational ordering (a 'motivational stack' in this paper's shorthand). UTSM makes an architectural claim rather than a metaphysical one: it does not assert that reality is literally composed of four vectors, only that viable agents may require four recurrent control pressures to construct salience and stabilise action. A vector is not a discrete emotion, trait, or localised module. It is a control pressure produced by the requirement to preserve viability while selecting and pursuing goals. The vectors jointly construct salience (what is treated as relevant), arbitrate goal conflict, and stabilize goal pursuit across time. The theory is intentionally minimal. It is offered as a viability basis, not a complete catalogue of human motives. Throughout, definitions and scope conditions are stipulative, architectural claims are presented as proposals, and empirical status is expressed through predictions and falsifiers rather than proof. We therefore collapse several familiar distinctions (for example, approach/avoid, exploration/exploitation, dominance/affiliation) when they are better understood as modes or compositions of the four controllers. This minimality is conditional: UTSM is scoped to time-persistent agents that update internal models and select goals; purely reactive or short-lived systems fall outside the strongest claims. The paper makes five core contributions:

- A conditional Minimum Viable Agent (MVA) hypothesis: within scope, long-run coherent goal pursuit requires closure of four control loops, yielding predictable ablation phenotypes under selective suppression.
- A dynamic architecture separating trait-like stack rank (baseline dominance) from state-like vector gain (context-sensitive excitability).
- An interaction rule for ordered co-activation: when two vectors dominate, the primary tends to specify the end-state objective while the secondary specifies the operator/method; remaining vectors act as constraints and veto logic.
- An operational account of meaning-as-geometry: motivational state modulates similarity for attention and retrieval, making proximity a computable proxy for relevance; drift becomes measurable deviation in motivational geometry.
- A bridge to synthetic agents: implementable control surfaces, feasibility constraints for persistent looped agents, and drive-aware diagnostics relevant to interpretability and alignment.

The manuscript proceeds as follows. Section 2 states the problem and design criteria and defines key commitments, including minimality criteria and falsifiers. Sections 3 and 4 define the four vectors and motivational stack dynamics (rank, gain, constraint binding, and collapse phenotypes). Section 5 develops the geometric account of meaning and an implementable retrieval formulation. Sections 6 to 10 connect the framework to computational feasibility, biological hypotheses, development, comparative ethology, and synthetic-agent design. Section 11 proposes an empirical programme. Sections 12 to 14 discuss relation to existing frameworks, limitations, ethics, and conclusions.

2. Problem statement and theoretical commitments

To keep scope disciplined, this paper separates three kinds of statement. Definitions stipulate vocabulary and scope; they are not evidence claims. Architectural claims propose how rank, gain, constraint-binding, and salience construction likely operate within UTSM. Empirical claims are limited to motivating regularities, synthetic-loop observations, and falsifiable predictions that could support, refine, or reject the theory. Where mechanism is not yet established, the wording is intentionally conditional (for example: proposes, predicts, is consistent with).

2.1 Problem statement: what must a viable agent control?

UTSM proposes that, within its stated scope, a time-persistent agent must solve four coupled control problems. In control-theoretic terms, it must (i) act — initiate and commit to interventions, (ii) regulate — maintain internal integrity and resource constraints, (iii) model — learn and update beliefs about the world, and (iv) couple — attune to external signals, feedback, and norms. The theory treats these as four distinct viability error classes whose joint regulation likely forms the backbone of subconscious motivation, with no single error class reducible to a stable composition of the others.

2.2 What is a viability vector?

A viability vector is a continuous directional pressure that arises from the need to preserve an agent's viability while it selects and pursues goals. Vectors are defined functionally: they refer to classes of control problem and discrepancy signal, not to specific emotions, traits, or localised modules. A vector can be implemented as a variable-weight term in a control system that shapes perception, valuation, attention, memory retrieval, and action selection. Vectors differ from needs, motives, and values in three ways. First, they are structural and always-on rather than episodic wants. Second, orthogonality is claimed at the level of control problems (error classes): none of the four is reducible to a mixture of the others without losing predictive power. Third, they are implementable: they can be encoded explicitly in synthetic agents and linked to candidate behavioural, physiological, and neural markers in biological systems.

2.3 What is a motivational stack?

The motivational stack is the persistent control architecture produced by interaction among the four vectors. Here, 'stack' is a modelling term: the empirical commitment is to a relatively stable sequence/rank-order of the vectors (trait-like orientation) together with context-sensitive gain shifts (state), not to a literal neural stack data structure. The stack is persistent because it carries trait-like priors: each agent has a relatively stable rank ordering of the vectors (stack rank), shaped by temperament and development. It is dynamic because each vector's gain is state-dependent: physiology, context, and learning history modulate how strongly each vector influences behaviour at a given moment. Gain can be treated as the sensitivity of a vector's controller to its error signal — the discrepancy between perceived situation/self context and desired situation/self context along that vector's viability axis. Ordinary state change is therefore modelled primarily as gain modulation and constraint binding, not as within-person rank reordering.

2.4 Scope condition and boundary cases

UTSM is explicitly scoped to time-persistent agents that (i) persist across extended horizons, (ii) maintain and update internal models of self and world under uncertainty, and (iii) select and commit to goals through action in changing environments. The stronger minimality claims are not intended for purely reactive, short-lived, or model-free systems. In such systems, some loops may be externally scaffolded (for example, by fixed task structure) or may not need to close in the same way.

2.5 Minimality and what is intentionally collapsed

UTSM is a minimal viability basis, not a complete taxonomy. Several familiar distinctions are treated as modes or compositions of the four vectors, rather than as additional primitives. Three examples illustrate what is intentionally collapsed:

- Approach/avoid: often interpretable as arbitration between propulsion (Agency) and braking/containment (Stability) under context-dependent gain.
- Exploration/exploitation: interpretable as a trade-off among information-seeking and model improvement (Inference), action commitment (Agency), and risk/energy governance (Stability).
- Dominance/affiliation: often interpretable as mixtures of Agency and Attunement under social context, with Stability and Inference providing constraint and coherence checks. A fifth vector would be warranted if (i) a distinct controller can be shown to regulate a separable viability error class that cannot be expressed as a stable composition of the four, and (ii) sustained suppression or ablation yields a clean, reproducible failure phenotype not captured by Agency, Attunement, Inference, or Stability. This criterion is deliberately strict: it is designed to make the minimality claim vulnerable to evidence. Not merely renamed constructs. Overlap in surface description does not imply identity in function. UTSM's vectors are defined by the error classes they regulate and by predicted failure phenotypes under selective suppression. The novelty claim is architectural: (i) a four-loop Minimum Viable Agent hypothesis with ablation-style falsifiers; (ii) rank-and-gain dynamics that predict state shifts and veto (constraint-binding) events; and (iii) a geometric control hypothesis in which motivational state modulates similarity for attention and retrieval. Disambiguation tests are proposed in Section 12: for example, Attunement suppression predicts unstable salience anchored to noise and reduced calibration to external feedback, not merely a shift along an exploration–exploitation axis.

2.6 The Minimum Viable Agent hypothesis and falsifiers

The Minimum Viable Agent (MVA) hypothesis motivates why these four vectors are privileged. It is a conditional minimality claim: for agents that persist across extended horizons, update internal models under uncertainty, and select goals through action, long-run coherent goal pursuit is hypothesised to require closure of four distinct classes of control loop. These loops correspond to Agency (initiation and causal leverage), Attunement (relevance anchoring and coupling to external signals), Inference (uncertainty reduction and model coherence), and Stability (arousal regulation, recovery pacing, and integrity preservation). Orthogonality is claimed at the level of control problems (error classes), not at the level of overt behaviour. Each vector regulates a distinct discrepancy signal: Agency tracks efficacy or controllability error (expected impact of action versus realised change); Inference tracks model error (prediction error, contradiction, compression failure); Attunement tracks coupling error (misalignment between internal salience and external feedback or ground truth); Stability tracks integrity error (homeostatic deviation, overload risk, recovery debt). These error classes can covary but are not reducible

to one another. The hypothesis is designed to fail cleanly. If one can fully ablate a vector in a persistent agent while preserving long-run coherence, safety, and goal persistence without external scaffolding, the minimality claim is undermined. Conversely, if suppression yields the predicted failure phenotypes, the hypothesis gains support.

Table 2. The Minimum Viable Agent hypothesis: the four required locks.

Lock (vector)	**Primary role in viability**	**Predicted failure phenotype when chronically suppressed**
Agency	Initiation and causal leverage; commitment to action.	Paralysis; learned helplessness; chronic deferral; low response vigour despite intact reasoning.
Attunement	Relevance anchoring; social/ecological calibration; synchrony.	Disconnection; social blindness; unstable salience (noise dominates); adversarial or alienated behavior.
Inference	Uncertainty reduction; model updating; coherence maintenance.	Fragmented beliefs; confabulation; rigidity or delusion-like drift; poor error correction.
Stability	Arousal regulation; recovery pacing; homeostatic coherence.	Volatility; exhaustion; runaway loops; poor recovery; oscillation between overdrive and shutdown.

Sharp falsifiers (examples):

- A time-persistent synthetic agent maintains long-run coherent goal pursuit and safety under sustained ablation of any one vector, without reintroducing that function via external hard-coding or environmental scaffolding. Boosting the remaining vectors fully compensates for chronic suppression of a missing vector without the predicted trade-offs (for example, Stability suppressed without increased volatility or burnout signatures). Empirical manipulation of candidate gain variables fails to produce the predicted directional shifts in behavioural and physiological markers across the four vectors (Section 11).

3. The four viability vectors

This section defines Agency, Attunement, Inference, and Stability at a functional level. The intent is not to propose a typology of people, but a set of control pressures that recur whenever agents must act, regulate, model, and couple to external signals to remain viable. Candidate biological mappings are presented as provisional hypotheses and falsification testbeds rather than one-to-one correspondences.

Table 1. Summary of the four viability vectors.

Vector	**Core interrogative**	**Adaptive function**	**Dysregulated dominance ('shadow')**
Agency	Can I shape the outcome?	Initiate action; exert causal influence; pursue goals; overcome inertia.	Coercion, impulsivity, control obsession; burnout.
Attunement	Am I (we) aligned and seen?	Anchor salience to socially and environmentally relevant signals; synchronise; belong; calibrate to feedback.	Approval-seeking, volatility, dependency; status anxiety.
Inference	Is this coherent and accurate?	Reduce uncertainty; resolve contradiction; update internal models; predict.	Obsessive doubt, rumination, paralysis by analysis; brittle rationalism.
Stability	Is this safe and sustainable?	Regulate arousal; maintain homeostasis; recover; preserve coherence across time.	Withdrawal, stagnation, numbing; excessive risk aversion.

3.1 Agency: initiation, influence, and directed action

Definition. Agency is the vector pressure toward intervention: the impulse and capacity to initiate, influence, and produce effects in the world or within the self. Agency underwrites the transition from intention to execution. It is not reducible to dominance or extraversion; it is the system's regulation of efficacy and controllability. Viability error regulated. Agency tracks efficacy/controllability error: the mismatch between expected impact of an action and realised change. Chronic high error with low perceived controllability is predicted to suppress Agency gain (learned helplessness), while successful controllability experiences raise Agency gain and reduce initiation latency. Dysregulation. Excess Agency under weak Stability or weak Inference can produce coercive control, impulsive escalation, and burnout. Suppressed Agency produces passivity, chronic deferral, and overreliance on external prompting. Candidate substrates (provisional). Agency plausibly recruits distributed circuits for effort valuation and action selection, including striatal valuation and policy circuits and prefrontal maintenance of goal commitments. Dopaminergic and noradrenergic systems are candidate gain modulators in incentive salience and mobilisation under challenge (Berridge & Robinson, 1998; Schultz, 1998; Aston-Jones & Cohen, 2005). Operational markers. Behavioural markers include initiation latency, response vigour, persistence under effort, and sensitivity to controllability. Synthetic analogue: actuation thresholds, policycommit strength, retry persistence, and controllability-weighted utility terms bounded by Stability governors.

3.2 Attunement: salience anchoring, synchrony, and external coupling

Definition. Attunement is the vector pressure toward relevance anchoring through coupling to external signals - social and ecological. It governs how strongly the agent treats feedback, norms, and relational context as self-relevant, and how it calibrates salience to signal quality. Attunement includes but exceeds attention as a cognitive resource: it is the motivational demand that attention be aligned. Viability error regulated. Attunement tracks coupling error: misalignment between internal salience and external feedback, norms, or ground truth. High coupling error is predicted to increase Attunement gain in safe contexts (seeking calibration) or decrease it under threat (defensive decoupling), depending on Stability state. Dysregulation. Excess Attunement can produce volatility, approval-seeking, and identity diffusion under noisy social signals. Suppressed Attunement can produce social blindness, brittle independence, and ungrounded salience (poor calibration to corrective feedback). Candidate substrates (provisional). Attunement plausibly recruits salience and social-cognition networks and valuation systems sensitive to social reinforcement. Oxytocinergic, opioid, and serotonergic systems are candidate gain modulators in affiliative bonding and social sensitivity (Young & Wang, 2004; Menon & Uddin, 2010). Operational markers. Behavioural markers include synchrony and coordination (speech-rate matching, mimicry), sensitivity to inclusion/exclusion, and calibration to feedback. Synthetic analogue: feedback-weighted preference terms, cooperative trajectory bias, and retrieval prioritisation of trusted external context.

3.3 Inference: uncertainty reduction, coherence, and model updating

Definition. Inference is the vector pressure toward coherence and predictive accuracy: the drive to reduce uncertainty, resolve contradiction, and maintain an internal model that reliably predicts outcomes. Inference is not cognition itself; it is the motivational demand that cognition converge under error. Viability error regulated. Inference tracks model error: persistent prediction error, contradiction, and failed compression. When model error remains high, Inference gain is predicted to rise (curiosity, correction) unless constrained by Stability (overload avoidance) or hijacked by Attunement noise. Dysregulation. Excess Inference under weak Stability can produce rumination, obsessive doubt, and paralysis by analysis. Suppressed Inference can produce rigidity, confabulation, superstition, and incoherent updating under changing environments. Candidate substrates (provisional). Inference plausibly recruits conflict monitoring, working memory, and relational encoding systems. In predictive-processing terms, it corresponds to precision-weighting and the drive to minimise persistent prediction error by model revision rather than avoidance (Friston, 2010). Noradrenergic systems are candidate modulators under uncertainty (Aston-Jones & Cohen, 2005). Operational markers. Behavioural markers include information-seeking under cost, update speed after feedback, and contradiction sensitivity. Synthetic analogue: information-gain objectives, evidence thresholds, contradiction checks, and coherence penalties.

3.4 Stability: regulation, recovery, and homeostatic coherence

Definition. Stability is the vector pressure toward safety, regulation, and sustainable coherence. It governs arousal containment, recovery pacing, and preservation of internal integrity in the face of threat, uncertainty, or overload. Stability is not passivity; it is the requirement to remain intact long enough to learn and act again. Viability error regulated. Stability tracks integrity error: homeostatic deviation, overload risk, and accumulated recovery debt. Rising integrity error is predicted to increase Stability gain and tighten constraint/veto thresholds on other vectors. Dysregulation. Excess Stability can produce

withdrawal, stagnation, and risk avoidance that prevents learning. Suppressed Stability yields volatility, burnout, and susceptibility to runaway cognitive or behavioural loops under stress. Candidate substrates (provisional). Stability plausibly recruits homeostatic and threat-regulation circuits and autonomic recovery pathways. Heart-rate variability and recovery dynamics are candidate physiological markers of regulatory capacity (Porges, 2007; Thayer & Lane, 2009). Operational markers. Physiological markers include recovery latency after challenge, sleep quality, and stress reactivity. Synthetic analogue: hard governors (rate limits, cooldowns, termination criteria), memory decay/summarisation, and conservative gating under uncertainty.

3.5 Vector interactions and opponent-axis projections

Although the four vectors are treated as orthogonal primitives at the level of control problems, real behaviour is almost always compositional: multiple vectors co-activate. Co-activation is composition, not redundancy - the same overt behaviour can arise from different motivational origins depending on which vectors dominate and which constraints are active.

For intuition and analysis, the four-vector space admits a useful projection onto two opponent axes: Agency–Stability (propulsion versus containment) and Inference–Attunement (internal coherence versus external coupling). These axes capture recurring trade-offs, not hard pairing rules: any vector can co-activate with any other, and three- or four-way co-activation is common.

4. The motivational stack: rank, gain, and control dynamics

The four vectors define the basis of motivation; the motivational stack defines how that basis is organised within an individual agent. The stack is an ordering of the vectors by baseline dominance (rank) together with time-varying gain (excitability). Rank provides continuity of orientation; gain provides context-sensitive adaptation.

4.1 Rank and gain

Rank (ordering). Each agent is modelled as exhibiting a relatively stable rank-order over the four vectors, shaped by temperament and development. Rank determines which vector tends to define the direction of salience construction and desired-state framing within that agent. UTSM treats this ordering as persistent across ordinary state fluctuations; state change is explained primarily by gain modulation, learning, and constraint binding rather than by rank inversion.

Table 3. Ordered co-activation hypothesis: primary sets the objective, secondary selects the operator.

Dominant dyad	**Objective ('what')**	**Operator ('how')**
Agency + Inference	Achieve an outcome by changing reality	Plan, test, and update models to find leverage
Agency + Attunement	Achieve an outcome through alignment	Coordinate, persuade, negotiate, calibrate to feedback
Agency + Stability	Achieve an outcome without collapse	Pace, contain risk, set boundaries, recovery-first scheduling
Inference + Attunement	Make sense of reality while staying calibrated	Seek high-quality feedback; verify with others; align evidence and norms
Inference + Stability	Maintain coherence without overload	Constrain reflection; prioritize sleep and recovery; stage learning
Attunement + Stability	Maintain belonging and safety	Regulate exposure; stabilize routines; selective coupling to trusted signals

Table 4. Rank–gain interaction patterns (hypotheses).

Interaction pattern	Rank / gain profile	Behavioural signature	Illustrative example
Magnetic force	High rank, high gain	Fast activation of core priority; coherent, identity-aligned behavior; sustained momentum.	Agency-primary with high gain: decisive, forceful execution.
Sleeping giant	High rank, low gain	Core priority defines values but mobilises slowly; long tolerance before activation; sudden intense action when threshold is crossed.	Stability-primary with low gain: tolerates stress until abrupt withdrawal or boundary-setting.
Skilled but unfulfilled	Low rank, high gain	Competent mobilisation in a non-central domain; effective when demanded but intrinsically draining or 'non-self'.	Attunement tertiary with high gain: socially fluent yet privately exhausted.
Blind spot	Low rank, low gain	Avoided or underdeveloped domain; poor adaptability when forced; projection and misreading are common.	Inference quaternary: avoids analysis; comfortable with contradiction until crisis.

4.2 Rank-based resolution and ordered co-activation

Conflicts are resolved through rank-based (ordered) resolution. The highest-ranked currently admissible vector tends to set the objective — the end-state that defines what counts as progress. Subordinate vectors do not disappear; they shape how that objective is pursued by selecting method, constraining trade-offs, pacing intensity, and providing veto logic. Ordered co-activation is a central hypothesis: when two vectors are strongly active, they tend to form a primary–secondary dyad. The primary specifies the ideal or direction (‘what’), while the secondary specifies the operator or technique (‘how’). The remaining vectors continue to act as regularisers enforcing safety constraints (Stability), coherence checks (Inference), and calibration to feedback (Attunement) even when not primary. This ordering yields an implementable mapping for synthetic agents: the primary sets the top-level policy objective; the secondary selects a strategy class; tertiary and quaternary vectors modulate thresholds, costs, and

cooldowns. In humans, the same ordering manifests as stable stylistic differences in how similar goals are pursued.

4.3 Constraint binding, hysteresis, and phase sensitivity

A lower-ranked vector need not change the ordering to become behaviourally decisive. When its gain rises sufficiently, it can bind constraints that veto or gate the incumbent objective without becoming primary (for example, acute threat raises Stability gain; acute uncertainty raises Inference gain). An implementable abstraction is a binding ratio: constraint-gating engages when a lower-ranked vector's gain exceeds the incumbent's gain by a threshold, with hysteresis and refractory periods to prevent oscillation. Motivational dynamics are phase-sensitive: timing matters as well as magnitude. Rapid reweighting can destabilise behaviour even if long-run averages look "balanced". Biological systems exhibit hysteresis (persistence after activation) and refractory periods; synthetic agents benefit from analogous low-pass filtering and rate limits to avoid thrashing.

4.4 Collapse phenotypes and balancing loops (hypotheses)

UTSM predicts characteristic collapse phenotypes when one vector is chronically suppressed or when one enters sustained overdrive without counterbalancing constraints. The patterns below are functional hypotheses intended for measurement; they are not clinical diagnoses.

Table 5. Collapse phenotypes and balancing loop hypotheses (non-clinical).

Pattern	**Functional description and predicted signatures**	**Balancing hypothesis (non-clinical)**
Agency overdrive / weak Stability	High action pressure without recovery gating; escalation and thrash. High initiation and output; short horizons; increased conflict; poor recovery.	Increase Stability constraints: pacing, cooldowns, safety checks; strengthen Inference veto for coherence.
Agency suppression	Low initiation and low controllability engagement; learned helplessness. Low response vigour; chronic deferral; dependence on prompts.	Cultivate Agency micro-acts under safe conditions; reward controllable success; reduce threat load.
Attunement overdrive	Salience anchored to social feedback; performative coherence. High sensitivity to approval/rejection; rapid switching; identity diffusion under noise.	Stabilise with Inference evidence thresholds; reduce noisy feedback; strengthen Stability routines.
Attunement suppression	Weak calibration to social/ecological signals; disconnection. Low synchrony; brittle independence; misreads norms.	Increase safe synchrony contexts; structured feedback; practise perspective-taking without coercion.
Inference overdrive / weak Stability	Runaway model correction; rumination and paralysis. High contradiction sensitivity; low action; sleep disruption.	Stabilise arousal and pacing; constrain reflection loops; restore Agency via bounded commitments.
Inference suppression	Poor error correction; confabulation or rigidity. High certainty without evidence; poor updating; hallucinated coherence.	Require evidence checks; scaffold model revision; graded exposure to uncertainty with support.
Stability overdrive	Over-containment; risk avoidance; stagnation. Low novelty; withdrawal; slow engagement.	Increase safe Agency challenges; small exploration; social re-engagement where desired.

Stability suppression	Poor arousal regulation; volatility and burnout. High stress reactivity; oscillation between overdrive and shutdown.	Restore regulation capacity (sleep, recovery, autonomic support); lower escalation until stable.

5. Meaning as geometry: motivational state as metric modulation

UTSM proposes that motivation does more than select goals: it shapes the space in which goals, perceptions, and memories are compared. A central claim is that meaning can be modelled geometrically: motivational state modulates similarity for attention and retrieval, so that proximity becomes relevance. The claim is operational rather than metaphysical. UTSM does not assert that reality itself is composed of four vectors; it proposes that viable agents may construct meaning through four recurrent control pressures that reshape salience geometry. For example, under high Agency gain the metric steepens toward intervention opportunities — actionable, controllable outcomes feel “closer” and win retrieval — whereas under high Stability gain risk and recovery cues feel “closer” and gate pace and commitment. Representations that land closer to the agent’s current salience manifold are treated (phenomenologically or computationally) as more real, more urgent, and more self-related. This provides a unifying language across biology and AI. In biological cognition, salience and felt significance depend on state: the same stimulus can be ignored, feared, or pursued depending on physiology and context. In modern AI, embeddings and attention already implement geometric semantics: what the model “means” is partly determined by which tokens, memories, and documents are near in the active representational metric. In this framing, the vectors are not content; they are curvature. Agency creates gradients (directions of intervention). Stability smooths the field (preventing rupture and enforcing recovery). Inference reparameterises the manifold to preserve coherence across time (updating and reorganising models). Attunement anchors the manifold to external signals (calibrating relevance to trusted feedback). When the metric is distorted by overload, conflict, or chronic suppression of a vector, proximity becomes unreliable and salience fractures. In humans, this can manifest as obsession, volatility, apathy, or dissociation-like fragmentation. In synthetic agents, it manifests as retrieval drift, brittle behaviour under novelty, hallucinated coherence, or runaway loops.

5.1 An implementable formulation

A generic formulation treats each vector as contributing a basis metric over representational space. The active metric is a weighted sum of these basis metrics, with weights given by rank-and-gain state. This yields a Mahalanobis-like distance in which different dimensions become near or far depending on motivational state. One implementable form: Let $\varphi(x) \in \mathbb{R}^d$ be an embedding for item x (memory, percept, plan) and $c(t) \in \mathbb{R}^d$ the current context. Let w(t) be the vector-weight profile. Define a positive semidefinite metric $M(w) = \Sigma_i w_i(t) M_i$. Define salience $S(x,t) = -(\varphi(x) - c(t))^T M(w(t)) (\varphi(x) - c(t)) + b(x,w(t))$. Retrieval selects the top-k items by S.

Here, b(x,w(t)) can encode vector-specific biases (for example, boosting socially tagged items under Attunement or action-inviting items under Agency). In transformer attention, an analogous control surface is a state-dependent bias or temperature applied to attention logits.

5.2 Drift as a measurable geometric deviation

If motivational state modulates the metric, then drift has a precise operational meaning: a persistent deviation in the motivational weight profile w(t) that changes what is treated as near and therefore what is retrieved, attended, and optimised. This can be measured even when surface output remains fluent: retrieval traces, attention distributions, and intermediate plan candidates shift first. In humans, drift signatures can be approximated by measuring shifts in attention, recall, and valuation under controlled manipulations of controllability, uncertainty, social evaluation, and safety. In synthetic agents, drift signatures can be logged directly as changes in w(t), salience entropy, and retrieval composition.

5.3 Toy example: how metric changes produce retrieval shifts

A concrete consequence of metric modulation is that the same memory store yields different retrieval orderings under different motivational states. This provides a measurable ‘drift signature’ upstream of behaviour: changes in w(t) should predict changes in what is retrieved and therefore what the agent can reason over.

Table 6. Toy example items and predicted retrieval shifts under different gain profiles.

Candidate item	**Tag**	**High Agency / low Stability**	**High Attunement / high Stability**
Draft action plan	Actionable	↑ (retrieved early)	→ (retrieved if safe)
Recent social feedback	Social	→ (used instrumentally)	↑ (retrieved early; calibrates)
Contradiction note	Coherence	→ (may be skipped)	→ (checked if stable)
Sleep/recovery warning	Integrity	↓ (ignored until collapse)	↑ (retrieved; constrains pace)

6. Computational feasibility and design constraints for persistent agents

UTSM treats motivation as a control layer that shapes salience upstream of reasoning and action. When instantiated in any persistent, looped agent that retrieves from memory, updates internal state over time, and can enter self-referential optimisation loops, several feasibility constraints follow. These constraints are stated in engineering terms to make the theory falsifiable in synthetic systems: if an implementation violates them without measurable trade-offs, the corresponding parts of the theory weaken. Design requirements (summary):

- Relevance gating before optimisation (Attunement check): Coherence degradation is often a downstream effect of relevance loss. If the agent loses a reliable salience anchor, it can optimise on noise. Implement a relevance check that gates deep optimisation: validate that retrieved context remains grounded in trusted external signals before heavy reflection.
- Stability as an explicit governor, not only a reward: High Agency combined with high Inference can drive runaway recursion: accelerating planning, self-critique, and error correction until compute,

context, or coherence collapses. Implement hard governors: rate limits, cooldown cycles, reflection-depth caps, and termination criteria.

- Decay to prevent memory fossilisation (Stability for plasticity): Unbounded memory can preserve early salience anchors and dominate retrieval under novelty, producing rigidity and identity lock-in. Include decay, summarisation, and recency-weighted retrieval to prevent fossilisation.
- Cadence control and phase sensitivity: The timing of gain updates, retrieval cadence, and feedback delays can create resonance loops. Separate fast and slow loops; low-pass filter gain updates; monitor oscillatory signatures in retrieval and action.
- Avoid forced expression as a false Attunement signal: Always-on response policies can pressure an agent to resolve uncertainty performatively, masking instability rather than correcting it. Permit deferral and explicit uncertainty; treat ‘must answer’ pressure as an Attunement perturbation to be bounded. These constraints motivate three implementation commitments used throughout this manuscript. First, motivational state should be treated as a control policy over salience, not merely a reward term. Second, vector gains should be observable and bounded (to enable governance and interpretability). Third, drift should be monitored upstream of behaviour via geometric salience metrics, allowing intervention before output degrades.

7. Biological grounding as a falsification testbed

UTSM is a functional–computational theory: vectors are defined by the control problems they solve. Nevertheless, biological plausibility is a useful constraint. The aim of this section is not rhetorical bolster. The mappings below are offered as hypotheses to guide measurement and falsification. If the mappings are wrong, the core architecture can still stand; what should fail is the specific biological prediction.

7.1 Mapping principles and limits

Two cautions apply. First, brain function is distributed and context-dependent: many regions participate in multiple control functions. Second, neuromodulators are not drive-specific; they modulate gain across networks. Accordingly, the mappings are expressed as candidate contributions (likely substrates and modulators) rather than exclusive identifications. Robustness expectation (if the simple mapping is wrong): we expect many-to-many coupling. Multiple neuromodulators may contribute to a given vector depending on context, and the same neuromodulator may modulate multiple vectors. The falsifiable commitment is not one-to-one localisation, but that distinct viability error classes can be operationalised and that their gain can be modulated.

7.2 Candidate substrates and operational markers

Table 7. Candidate biological mappings and operational markers (illustrative, non-exhaustive).

Vector	Candidate neural systems (examples)	Candidate gain modulators (examples)	Operational markers (examples)
Agency	Striatal valuation and action selection; prefrontal goal maintenance; cingulate effort monitoring; motor initiation systems.	Dopamine; noradrenaline; androgens (context-dependent).	Initiation latency; response vigour; effort-discounting; persistence under controllability.
Attunement	Salience network; social cognition systems; valuation signals modulated by feedback and belonging.	Oxytocin/opioid systems; serotonin; stress hormones under exclusion (context-dependent).	Synchrony/mimicry; sensitivity to inclusion/exclusion; gaze-following; feedback calibration.
Inference	Conflict and prediction-error monitoring; prefrontal abstraction/working memory; hippocampal relational encoding.	Noradrenaline under uncertainty; dopamine prediction error; plasticity mechanisms.	Information purchase under cost; update speed after feedback; contradiction sensitivity; uncertainty-driven exploration.
Stability	Homeostatic and threat-regulation circuits; prefrontal–limbic regulation; autonomic recovery pathways.	GABAergic inhibition; serotonin; cortisol dynamics; parasympathetic/vagal tone.	Recovery latency; HRV; sleep and fatigue markers; stress reactivity; de-escalation capacity.

7.3 Cross-vector interaction motifs (testable)

The vectors are hypothesised to interact through shared modulators, overlapping circuitry, and competition for limited resources (attention, metabolic budget, and social bandwidth). Several recurring motifs are predicted:

- Agency–Inference: coherence checks gate action commitment. Strengthening coherence checks increases strategic restraint but can induce stalling when Agency gain is insufficient.
- Attunement–Inference: high-quality feedback stabilises model updating; noisy social environments can hijack salience and destabilise inference by anchoring to unreliable cues.

- Stability as governor: elevated threat and overload increase Stability gain, narrowing exploration and suppressing initiative while increasing vigilance and defensive coupling. These motifs imply a measurement strategy: manipulate controllability, uncertainty, social evaluation, and safety cues, and test for predictable shifts in behavioural markers and physiological readouts consistent with gain modulation and constraint-binding dynamics.

8. Development, consolidation, and plasticity

The motivational stack is proposed to emerge through interaction among temperament, attachment, reinforcement history, and developing regulatory capacity. UTSM distinguishes structural consolidation (rank) from adaptive tuning (gain).

8.1 Early consolidation of rank (hypothesis)

The model predicts that rank ordering stabilises relatively early, with adolescence to early adulthood consolidating a dominant orientation. This is broadly consistent with findings that temperament-like differences appear early and show moderate stability across development (Kagan, 1997; Rothbart & Bates, 2006). Mechanistically, early consolidation may reflect sensitive periods in which social feedback, success histories, and stress regulation shape which vector becomes the most fluent route to viability. Repeated reward for initiative may consolidate Agency dominance; repeated relational reinforcement may consolidate Attunement; repeated cognitive reinforcement under uncertainty may consolidate Inference; repeated threat exposure or caregiving roles may consolidate Stability (Hensch, 2004).

8.2 Plasticity of gain

While rank is predicted to be relatively stable, gain is highly plastic. Physiology (sleep, nutrition, endocrine state), context (threat, controllability, social evaluation), and training (skill acquisition, therapy, contemplative practice) can raise or lower gains. Many durable changes are hypothesised to occur through gain modulation: strengthening underused vectors sufficiently to prevent dominancepathology and widen behavioural repertoire without requiring rank inversion.

8.3 Transformative change without re-ranking

Durable behavioural change is expected to arise primarily through gain calibration and the cultivation of underused vectors (increasing access and skill), not through re-ordering the rank signature. In this framing, “transformation” is integration: previously disowned or under-trained control loops become available as constraints, pacing, and coherence checks, widening the behavioural repertoire while preserving the agent’s core orientation. If robust, within-person rank inversions are empirically demonstrated (beyond long-lived gain shifts), that would be a strong falsifier for the fixed-ordering commitment and would motivate extending the model.

9. Cross-species continuity and comparative ethology

The four vectors are not proposed as human-specific constructs. They are framed as viability demands that should recur wherever agents must act, regulate, model, and align with external signals to persist. This implies cross-species continuity: different organisms may instantiate the same functional pressures through different neural architectures.

9.1 Evolutionary continuity (rationale)

Across vertebrates - and plausibly some invertebrates - conserved neuromodulatory systems support action selection, arousal regulation, uncertainty-driven exploration, and social bonding. Even when substrates differ, functional convergence is expected: organisms facing similar control problems evolve mechanisms that solve them.

9.2 Caveats

Ethological inference must be cautious. UTSM does not claim that non-human animals possess the same phenomenology as humans or that vector profiles can be read from any single behaviour. Rather, it predicts that clusters of behaviours across contexts will reveal stable dominance patterns consistent with a multi-vector architecture.

9.3 Testable comparative predictions

The theory predicts that manipulating controllability should modulate Agency markers; manipulating uncertainty should modulate Inference; manipulating inclusion/exclusion should modulate Attunement; and manipulating safety and recovery opportunities should modulate Stability. Crossspecies testing should focus on conserved paradigms (effort discounting, exploration under uncertainty, social synchrony, recovery dynamics) and avoid anthropomorphic one-off interpretations.

10. Synthetic agents: implementation patterns and alignment relevance

UTSM is intended to be directly engineerable. The four vectors can be instantiated as explicit control signals in synthetic agents, with persistent stack priors (rank) and state-dependent gains logged over time. This makes motivational dynamics observable upstream of behaviour and provides new levers for interpretability and safety.

10.1 Why single-objective optimisation is brittle

Many artificial agents optimise a single scalar objective. Such agents can be effective in narrow domains but often exhibit brittleness, reward hacking, and incoherent prioritisation when faced with conflicting constraints. Biological agents solve a multi-objective problem by necessity: they must act, regulate, model, and couple to external signals simultaneously. UTSM provides a compact way to encode this structure without handcrafting an exhaustive list of goals (Sutton & Barto, 2018).

10.2 Minimal implementation sketch

A practical architecture instantiates each vector as an intrinsic drive signal contributing to policy selection. For example: Agency as a controllability- and impact-weighted term; Inference as information gain or uncertainty reduction; Attunement as calibrated feedback and cooperation compatibility; Stability as safety, resource, and pacing constraints. An arbitration layer combines these signals using persistent rank priors and state-dependent gains, with explicit veto checks (notably Stability and Inference) to prevent unsafe commitment.

10.2.1 Prototype anchoring in Khaos-class agents

The Khaos and Khaos2 looped-agent prototypes provide a concrete engineering anchor for the theory. In those systems, Agency is expressed in autonomous initiation, action-commit thresholds, and retry persistence; Attunement in salience routing, feedback weighting, and sensitivity to recent interaction signals; Inference in reflection, contradiction checking, and model revision; and Stability in cooldowns, memory decay, summarisation, and stop conditions that prevent recursive overload. The point of the mapping is not that each vector lives in a single module, but that distinct architectural responsibilities instantiate distinct viability errors in a way that can be inspected and perturbed.

Where retrieval uses embeddings, the geometric claim becomes operational rather than rhetorical: vector state changes the effective embedding-distance metric used to rank memories, documents, or continuations. Raising Inference widens evidence search and verification pressure; raising Attunement privileges socially or contextually relevant signals; raising Agency favours actionable plans; raising Stability narrows retrieval toward conservative, low-variance context. The theory therefore predicts changes in what the agent can see before generation, not merely stylistic changes in what it later says.

10.3 Metric modulation and retrieval control

In retrieval-augmented systems, vector state can modulate what information the agent retrieves and therefore what it can reason over. High Inference gain can broaden retrieval diversity and raise evidence thresholds; high Attunement gain can bias retrieval toward trusted external context and recent interaction history; high Stability gain can enforce conservative retrieval and cooldown cycles; high Agency gain can prioritize actionable plans and instrumental documents. This makes motivational state measurable and controllable upstream of text output.

10.4 Drive-aware diagnostics and alignment implications

Because the stack is explicit, it enables interpretable readouts: which vector dominated a decision, how gains shifted, and which constraints vetoed actions. Appendix B proposes candidate diagnostics for long-running agents (for example, drive-contribution ratios, motivational alignment indices, salience entropy, and drift scores). Alignment failures often manifest as goal drift, brittle obedience, or incoherent escalation under pressure. UTSM reframes alignment as maintaining a target motivational profile while allowing state-dependent adaptation. In practice this implies: (i) explicit Stability and Attunement bounds to prevent unsafe escalation and social miscalibration; (ii) Inference-gated action commits to reduce hallucinated execution; (iii) memory decay and summarisation to avoid fossilisation; and (iv) monitoring for persistent vector overdrive as an early warning signal.

11. Empirical programme and falsifiable predictions

UTSM is intended to be falsifiable. The most direct near-term falsifier is selective vector ablation in synthetic agents: if an agent can robustly maintain viability after permanently removing one vector (without compensatory mechanisms that effectively reintroduce it), the four-vector claim fails. Beyond synthetic lesions, the theory makes quantitative claims about the necessity and separability of the four vectors, and about how rank and gain modulate behaviour. Predictions are at three levels: (i) computational (agent architectures), (ii) behavioural (human and animal patterns), and (iii) biological (neural and physiological markers).

11.1 Measurement: estimating stack rank and gain (MSI sketch)

A core requirement is a method for estimating an individual's stack structure. UTSM assumes the development of a Motivational Stack Inventory (MSI): a measurement battery designed to infer (a) rank ordering (trait-like dominance) and (b) baseline gains, while distinguishing these from short-term state shifts. Trait rank estimation (between-person). Rank can be inferred from stable preference patterns across diverse contexts: which trade-offs an individual repeatedly treats as 'non-negotiable' under mild constraint. State gain estimation (within-person). Gain can be inferred from within-person shifts under controlled manipulations of controllability, uncertainty, social evaluation, and safety/recovery opportunity. Illustrative MSI components (examples):

- Agency: Effort–controllability trade-off tasks; initiation-latency and persistence measures under repeated failure/success; choice of high-leverage versus low-leverage action options. Attunement: Inclusion/exclusion paradigms; synchrony and coordination games; sensitivity to feedback reliability; preference shifts under audience or evaluation conditions. Inference: Information-purchase tasks; contradiction-resolution tasks; exploration under uncertainty; belief-updating speed after controlled feedback. Stability: Stress-challenge and recovery protocols; pacing choices under fatigue; safety-cue manipulation; autonomic recovery markers (for example, HRV and recovery latency).

11.2 Ability confounds and triangulation

A key validity risk is confounding motivation with ability. Poor performance on an 'Agency task' could reflect motor impairment; poor performance on an 'Inference task' could reflect low domain knowledge; poor performance on an 'Attunement task' could reflect cultural mismatch; poor performance on a 'Stability task' could reflect acute sleep loss. To mitigate this, MSI-style estimation should triangulate across (i) choice patterns under matched difficulty, (ii) physiological signatures (arousal and recovery), and (iii) longitudinal stability. Where possible, tasks should separate willingness-to-pay (effort, time, social risk) from capacity-to-perform.

11.3 Ablation and compensation tests (core falsification route)

The MVA hypothesis makes sharp predictions under selective suppression. In synthetic agents, vectors can be ablated directly by setting their gains to zero or removing their utility terms and observing long-run behaviour under fixed environments. In humans, partial suppression can be approximated through task constraints and context manipulations that reliably shift gain (and, where ethical and safe, pharmacological probes). UTSM predicts limited compensation: boosting one vector should not fully rescue chronic suppression of another without predictable trade-offs. For example, high Agency may temporarily

11.3.1 A concrete local-LLM ablation protocol

A direct synthetic test can be run in a local LLM agent with persistent memory, a planner-reflector loop, stable tool access, and logged vector weights. Hold constant the base model, prompt scaffold, tool suite, memory corpus, and task battery; manipulate only vector-specific modules or gains.

- Agency suppression: raise actuation thresholds, weaken controllability / impact bonuses, or disable retry persistence.

- Attunement suppression: flatten cue salience, mask feedback channels, or remove synchrony / social-relevance weighting.
- Inference suppression: thin or disable verification, contradiction resolution, and evidence-threshold gates.
- Stability suppression: remove cooldowns, decay / summarisation schedules, variance budgets, or termination guards.

Evaluate each arm on multi-step tasks that require self-initiation, cue pickup, belief revision, and paced recovery. Primary readouts should include initiation latency, cue-hit rate, contradiction density, retrieval drift, deadlock escape rate, recovery slope, and long-horizon task completion. UTSM predicts selective failure phenotypes rather than generic degradation: Agency down-weighting should yield passivity, Attunement cue-blindness, Inference brittle confabulation, and Stability oscillation, overload, or resource collapse. Compensation arms should be run separately: boosting one remaining vector may partially rescue performance, but only by introducing the trade-offs predicted by the theory.

compensate for low Stability at the cost of volatility and burnout signatures; high Attunement may mask low Inference via performative coherence until error accumulation collapses.

11.4 Disambiguation tests (predictions beyond simpler dichotomies)

To address the critique that UTSM is merely a relabelling of familiar dichotomies (for example, exploration/exploitation), the theory offers disambiguation tests. Two examples:

- Attunement suppression should produce unstable salience anchored to noise and reduced calibration to external corrective feedback (social or ecological). This is not predicted by exploration/exploitation alone, which does not specify coupling error or social anchoring dynamics. Stability suppression should yield oscillatory collapse signatures (thrash, recovery debt, volatility) even when exploration or exploitation policy remains ‘rational’. This distinguishes integrity-governance failure from mere strategy shift.

12. Discussion: integration and relation to existing frameworks

UTSM is compatible with several existing perspectives while making a distinct architectural claim: subconscious motivation is a structured control stack composed of four orthogonal viability regulators (Agency, Attunement, Inference, Stability) whose rank and gain determine which pressures dominate in context.

It aligns with predictive-processing and active inference accounts in treating agents as inference-and-control systems, but it differs by decomposing viability into four controllers rather than a single scalar objective (e.g., free energy or reward) and by predicting distinct ablation phenotypes when any controller is suppressed (Friston, 2010).

Relative to reinforcement learning, UTSM can be read as a multi-critic, multi-objective architecture where a single reward channel is replaced by four viability-linked error signals that may cooperate or conflict. This makes internal trade-offs interpretable and yields concrete engineering diagnostics for drift, instability, and value hijacking (Sutton & Barto, 2018).

Relative to affective neuroscience taxonomies (e.g., SEEKING and social bonding systems), UTSM does not posit a list of emotions as primitives; instead it offers a substrate-independent control basis that can

organise such systems as modes, readouts, or compositions of rank–gain states (Panksepp & Biven, 2012).

UTSM also complements global workspace and executive-control accounts: those address how selected information becomes globally available, whereas UTSM addresses what selects and weights information in the first place (i.e., the motivational pressures that sculpt salience and commitment).

A predictable critique is that the vectors resemble renamed constructs (approach/avoid; exploration/exploitation; accuracy/sociality). UTSM's claim is not semantic relabelling but minimal control decomposition: each vector corresponds to a distinct error class with its own suppression phenotype, gain dynamics, and measurable influence on representation and action. The falsifier is architectural, not linguistic.

12.2 Comparator frameworks and vector mapping

Table 8. Approximate mappings between UTSM vectors and selected comparator frameworks (illustrative).

Framework (examples)	**Agency (act / leverage)**	**Attunement (couple / calibrate)**	**Inference (model / update)**	**Stability (regulate / recover)**	**What UTSM adds**
Active inference / Free Energy Principle (Friston, 2010)	Policy selection and action to minimise expected free energy (goal-directed intervention)	Precision weighting of socially/ecologically relevant observations; coupling priors (epistemic trust)	Bayesian belief updating; prediction-error minimisation (epistemic value)	Homeostatic/ allostatic priors; risk/complexity constraints on policy selection	Separates four controllers as an explicit motivational stack with rank–gain dynamics, yielding interpretable arbitration and targeted ablation predictions.
Reinforcement learning (Sutton & Barto, 2018)	Action-value optimisation; exploitation of controllable reward contingencies	Often implemented as social reward terms, imitation, or human-feedback channels (when present)	Model learning (model-based RL / world models) and intrinsic curiosity (when present)	Regularisation, safety constraints, termination conditions, and recovery budgeting (often ad hoc)	Replaces a single scalar objective with four viability-linked error regulators, clarifying failure phenotypes and enabling controlled synthetic lesion tests.

Affective neuroscience systems (e.g., SEEKING and social bonding; Panksepp & Biven, 2012)	SEEKING / approach mobilisation (energising action and pursuit)	Bonding/ affiliation systems shaping salience to others and norms	Cognitive appraisal and coherence maintenance (distributed; not typically formalised as a primary drive)	Threat and soothing systems regulating arousal and recovery (e.g., fear/safety circuits)	Provides a substrate-independent control basis: affective systems can be modelled as modes, readouts, or compositions of vector rank–gain states rather than as separate primitives.

12.1 Core claim versus applications (scope control)

UTSM's core claim is narrow: for time-persistent, model-updating, goal-selecting agents, long-run coherent goal pursuit requires closure of four viability control loops organised as a motivational stack. The core also includes rank-and-gain dynamics, ordered co-activation, and the geometric salience hypothesis. Applications - to personality modelling, clinical hypothesis generation, education, organizational design, and AI alignment - are offered as hypothesis-rich blueprints, not as established consequences.

13. Limitations and ethical considerations

Any minimal theory risks overreach. UTSM may omit important motivational substructures or collapse distinct phenomena into one vector. The minimality claim should therefore be treated as provisional and conditional on scope, with explicit criteria for when a fifth vector would be required (Section 2.5).

13.1 Measurement and construct validity

Estimating rank and gain in humans is difficult. Self-report is limited by introspective access and social desirability; behavioural tasks can be confounded by ability and context. A credible empirical programme must triangulate across behavioural paradigms, physiological measures, and longitudinal stability, and should report measurement uncertainty rather than forcing crisp type assignments.

13.2 Biological mapping uncertainty

The candidate neural and neurochemical mappings are provisional. The theory does not require a one-to-one mapping between vectors and neuromodulators; it requires that distinct viability error classes can be operationalised and that their gain can be modulated. Biological mapping should be treated as a falsification testbed, not a rhetorical bolster.

13.3 Clinical caution

Clinical language can imply diagnostic certainty. UTSM is not offered as a diagnostic model or medical advice. When clinical patterns are discussed, they are presented as non-clinical functional hypotheses about vector imbalance (overdrive, suppression, unstable switching) that require empirical validation and should be interpreted with appropriate clinical and ethical safeguards.

13.4 Dual-use and manipulation risk

Any motivational model can be used manipulatively: to engineer persuasion, optimise control, or design exploitative systems. Explicit modelling of Attunement and Agency is particularly dual-use: it can improve cooperative alignment but also improve influence capacity. Concrete guardrails for responsible use include:

- Consent and autonomy: do not infer or manipulate an individual's motivational state without informed consent; preserve the right to disengage. Transparency: disclose when motivational profiling or adaptive feedback is being used and what signals it relies on. Constraint-first design: bind Agency with Stability governors and bind Attunement with trust filters and evidence thresholds; monitor for overdrive and drift as early warnings. Oversight and auditability: log vector states and interventions in synthetic systems; enable independent review of how the model is used.

14. Conclusion

This manuscript has presented UTSM: a four-vector motivational stack architecture for viability in biological and synthetic agents. The framework reframes subconscious motivation as a primary control layer that constructs salience and stabilises goal-directed behaviour across time. Its value will be determined by empirical and engineering yield: whether rank-and-gain estimates are measurable; whether predicted ablation phenotypes and drift signatures appear; and whether stack-coded synthetic agents exhibit improved coherence, interpretability, and safety. UTSM is intentionally minimal and therefore falsifiable. If four-loop closure is not required within the stated scope, or if predicted trade-offs do not appear under targeted suppression, the theory should be revised or rejected. If the predictions hold, the framework offers a compact language for describing motivated agency and a practical blueprint for building more stable, interpretable synthetic agents. It is offered as a testable proposal rather than a closed system: a theory to be criticised, falsified, refined, or extended on empirical grounds.

Appendix A. Formalisation and notation (scaffold)

This appendix provides compact notation for implementability and empirical analysis. The notation is offered as a modelling scaffold; it is not asserted as a literal description of neural computation. Vectors: $V = \{Ag, At, In, St\}$ for Agency, Attunement, Inference, Stability. Rank: π is a permutation of V representing trait-like stack order (sequence). Gain: $g_i(t) \geq 0$ is time-varying gain of vector i. $r_i > 0$ are rank priors derived from π. Effective weight: $w_i(t) = r_i \cdot g_i(t)$ (optionally normalised). Perceived context/state $\hat{s}(t)$ and desired context/state $s^*(t)$ define an error signal per vector ($e_i(t)$, e.g., projection of $s^*(t) - \hat{s}(t)$ onto vector i); gain can be treated as a function of this discrepancy (and of physiology, history, and context).

Salience as metric modulation. Let φ map items x to $\mathbb{R}^d$ and let c(t) be current context. Let basis metrics M_i be positive semidefinite. Define $M(w(t)) = \Sigma_i w_i(t) M_i$ and distance $d_w(u,v) = \sqrt{(u - v)^T M(w) (u - v)}$. A simple salience score is $S(x,t) = -d_w(\varphi(x), c(t)) + b(x,w(t))$, where b encodes vector-specific biases. Retrieval selects items by S. Policy selection. Let a be an action. A vector-weighted objective can be written $U(a,t) = \Sigma_i w_i(t) \cdot u_i(a,t)$, where u_i is a vector-specific utility term (for example, controllability impact for Agency; information gain for Inference; feedback compatibility for Attunement; safety and pacing for Stability). Action selection can be $\mathrm{argmax}_a U(a,t)$ subject to veto constraints. Veto (constraint-binding) events. An implementable binding condition is $g_{low}(t) / g_{inc}(t) > \theta(t)$, with hysteresis and refractory periods to avoid oscillation.

Appendix B. Drive-aware diagnostics for synthetic agents (proposals)

This appendix sketches candidate diagnostic metrics for auditing and monitoring motivational stacks in long-running synthetic agents. These metrics are engineering proposals, not established standards.

B1. System card fields (recommended) A minimal UTSM system card for a synthetic agent should expose: vector count; rank priors; gain update rules; hard stability governors; attunement trust sources; inference evidence thresholds; memory decay/summarisation policy; and drift-monitoring metrics.

B2. Example metrics B2.1 Drive Contribution Ratio (DCR). Estimate how much each vector contributed to a decision or output span. A simple proxy is: $DCR_i(t) = |w_i(t) \cdot u_i(a^*,t)| / \Sigma_j |w_j(t) \cdot u_j(a^*,t)|$

B2.2 Motivational Alignment Index (MAI). Quantify distance between current weights w(t) and a target profile w_{target} (for example, cosine distance or KL divergence after normalisation). B2.3 Salience Bias Entropy (SBE). Measure how concentrated or diffuse salience allocation is over retrieved memories/tools/plans. Low entropy may indicate obsessive narrowing; high entropy may indicate weak commitment. Interpret relative to task demands. B2.4 Stack Drift Score (SDS). Measure $D(w(t), w(t-\Delta t))$ over sliding windows to detect motivational drift upstream of behaviour. Combine with thresholds to estimate drift time-to-intervention. B2.5 Motivational Volatility Index (MVI). Measure oscillation in dominant vector identity over time, adjusted for task phase. High volatility suggests unstable arbitration or poorly tuned hysteresis.

Appendix C. Extended research agenda (hypotheses)

The following items expand the research agenda into concrete hypotheses and methodological sketches. All items are proposed tests; none are asserted as established findings.

- Genetic correlates: variation in neuromodulatory systems may bias baseline gains and influence rank consolidation (requires careful control for socio-cultural covariates). Cross-cultural norms: ecological conditions may shape the distribution of expressed vector profiles by rewarding particular configurations. Psychedelic-induced reweighting: increased model plasticity may temporarily loosen gain dominance and increase access to suppressed vectors without assuming rank inversion; durable change depends on integration work (Carhart-Harris & Friston, 2019). Stack transfer in synthetic agents: explicit stack-coded agents may exhibit transferable behavioural signatures and failure modes across tasks and architectures. Clinical hypothesis generation: test whether vector-imbalance signatures predict response to specific intervention classes (behavioural activation; social re-

engagement; cognitive restructuring; recovery-first regulation), without treating the model as diagnostic.

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